DECK 7 OF 10 · FRC JAVA CRASH COURSE

REV vs CTRE motors and controllers

What each one does on its own in 2026, and how to choose

WPILib 2026 · Java · Command-based

OVERVIEW

What this deck covers

Deck 2 used a SPARK MAX. Deck 6 ran PID on the roboRIO. This deck moves that loop onto the motor controller.

- Which motors work with which controllers.
- Why a fast loop on the controller beats a slow one on the roboRIO.
- The same arm in REVLib and in Phoenix 6.
- What costs money, and how to choose.

HARDWARE

The two vendors at a glance

	REV	CTRE
Motors on sale	NEO 2.0, NEO 550, NEO Vortex	Kraken X60, Kraken X44, Minion
Controllers	SPARK MAX, SPARK Flex	Talon FX, Talon FXS
Built in or separate	Separate. SPARK Flex docks straight onto a Vortex	A Kraken has a Talon FX built in. Minion uses a separate Talon FXS
Encoders	Through Bore Encoder V2, MAXSpline Encoder (wired or CAN)	CANcoder (CAN)
Other sensors	Color Sensor V3, 2m Distance Sensor	Pigeon 2 (IMU), CANrange (distance), CANdi (inputs on CAN)
Setup app	REV Hardware Client 2	Phoenix Tuner X
2026 software	REVLib 2026.0.5, with 2026 firmware	Phoenix 6 26.3.0, with 2026 firmware

Checked 2026-11-04. NEO V1.1 is discontinued and Falcon 500 is out of stock, but both are still legal if you own them.

HARDWARE

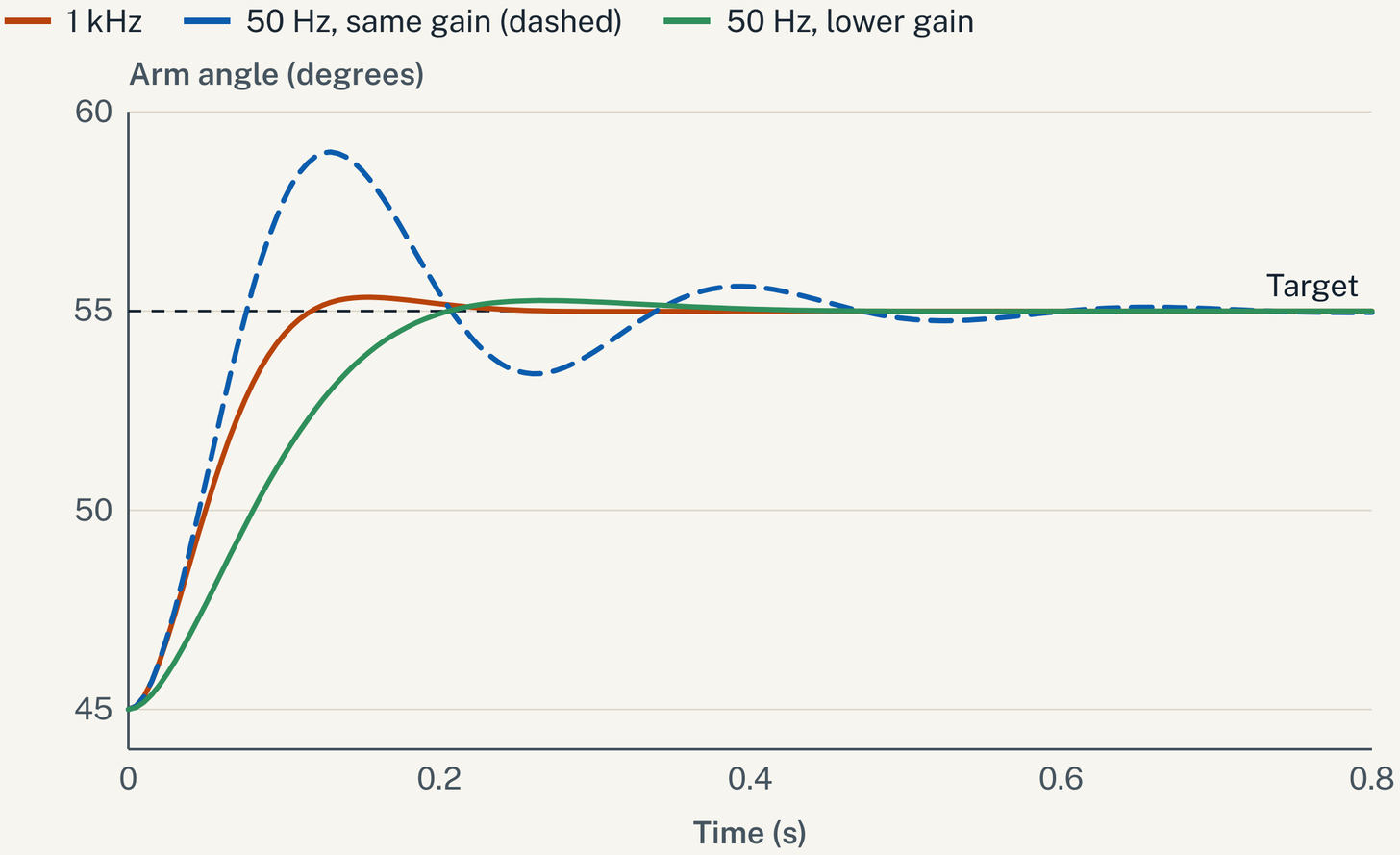
Which motor works with which controller

Motor	SPARK MAX	SPARK Flex	Talon FXS	Talon FX
NEO V1.1, NEO 550	Yes	With Flex Dock	Yes	No
NEO 2.0	Yes	With Flex Dock	Not listed by CTRE	No
NEO Vortex	With Solo Adapter	Docks directly	With Solo Adapter	No
Minion	Not documented	Not documented	Yes	No
Kraken X60, X44, Falcon	No	No	No	Built in
Brushed, such as CIM	Yes	With Flex Dock	Yes	No

Tell the controller what is attached. Brushed Mode with a brushless motor connected permanently damages the motor. A Talon FXS will not drive until you pick its motor in MotorArrangement.

CONTROL THEORY

Why a 1 kHz loop beats a 50 Hz loop



1. **Same gain at 50 Hz**
   The arm swings 4° past and takes 0.42 s to settle.

2. **Lower gain at 50 Hz**
   The swing goes away, but settling takes 0.17 s.

3. **Same gain at 1 kHz**
   No big swing, and it settles in 0.10 s.

ON THE CONTROLLER

Closed loop on the controller

	SPARK	Talon
Modes	Position, velocity, current, MAXMotion	Position, velocity, Motion Magic, in volts or duty cycle
Loop rate	Every 1 ms	1 kHz, so also every 1 ms
Feedforward	kS, kV, kA, plus an arbitrary term from your code	kS, kV, kA, plus a FeedForward term in the request
Gravity	kG for elevators. kCos with kCosRatio for arms	kG with GravityType. GravityArmPositionOffset shifts the arm angle
Gain slots	4, chosen in setSetpoint	3, plus simple gain scheduling on error
Arrived?	isAtSetpoint()	ClosedLoopError, and MotionMagicAtTarget for the profile
Following	config.follow(leader, invert), set on the follower	Follower request: copies the leader’s output type

Arm gravity works the same way on both: the sensor must read 0 when the arm is level.

ON THE CONTROLLER

Motion profiles on the controller

MAXMotion (REV)

Position: a trapezoid. Speed up, cruise, slow down.

You set cruise velocity, max acceleration and allowed profile error.

Drift too far off the plan and the SPARK makes a new plan from where it is.

Velocity: ramps to a speed at your max acceleration.

Motion Magic (CTRE)

Motion Magic: a trapezoid, or a smoother S-curve if you set jerk.

Velocity: ramps to a speed at your acceleration, with optional jerk.

Expo: an exponential curve from kV and kA that fits what the motor can do.

Dynamic: change the limits in every request. Needs Pro and a CANivore.

ON THE CONTROLLER

Limits and protection

	SPARK	Talon
Motor current	Smart Current Limit. Holds phase current. Default 80 A	Stator limit. Default 120 A, on
Battery current	No separate setting	Supply limit. Default 70 A, then 40 A after 1 s, on
Soft limits	Forward and reverse, in softLimit	Forward and reverse, in SoftwareLimitSwitch. Position targets get capped to them
Limit switches	Wired to the data port	Talon FXS: wired. Kraken X60: none, so use a CANcoder, CANdi or CANrange
Battery sag	voltageCompensation holds output steady	Voltage requests do this, up to what the battery gives

Brownout is the same on both. Below 6.3 V (roboRIO 1) or 6.75 V (roboRIO 2 default), the roboRIO sends CAN motor controllers an explicit disable.

BEST PRACTICE

Run the loop on the controller, limit every motor

> **If timing matters, run the loop on the controller. On every motor, set a current limit.**

The controller reads a wired sensor directly, every millisecond. And the 80 A SPARK default is twice the top of REV's suggested range for a NEO 550, which is 20 A to 40 A.

```
// SPARK
config.smartCurrentLimit(40);
// Talon FX
config.CurrentLimits.StatorCurrentLimit = 40;
config.CurrentLimits.StatorCurrentLimitEnable = true;
```

COST

Free features and paid features

REV: one tier
REVLib, Hardware Client 2, StatusLogger, MAXMotion and simulation. REV's docs list no paid software.

Phoenix 6: free
Volt and duty cycle closed loops. Motion Magic, Velocity and Expo. Swerve API. Tuner X. Simulation. Limited log export.

CANivore: hardware
A second CAN bus for CTRE devices, with CAN FD and CANivore timestamps.

Pro license
FOC and torque control on Talon FX. FOC gives about 15% more peak power. FusedCANcoder, SyncCANcoder, full log export, hoot replay.

Pro and CANivore
Timesync, device timestamps, Dynamic Motion Magic, full differential control.

Without Pro, a request with EnableFOC falls back to normal drive, but TorqueCurrentFOC will not run. A Season Pass covers 100 devices for one team number. Pro features are free in simulation.

TOOLS

Logging, tools and simulation

	REV	CTRE
Automatic log	StatusLogger: on by default, writes .revlog	Signal Logger: writes .hoot. Auto-starts on roboRIO 2, or roboRIO 1 with a USB drive
Read it	Opens directly in AdvantageScope	AdvantageScope or Tuner X. Export without Pro is a limited signal list
SysId	WPILib SysId	WPILib SysId, fed by Signal Logger
Setup app	Hardware Client 2: firmware, CAN IDs, run and plot motors, zero encoders	Tuner X: firmware, configs, plots, self test, log tools
Generates	No code generator. Templates for MAXSwerve and EasySwerve	Swerve project, Elevator subsystem, CANdle subsystem
Simulation	SparkMaxSim and SparkFlexSim run closed loop and MAXMotion	SimState classes for every Phoenix 6 device

SAME JOB, BOTH VENDORS

Arm setup: sensor, gains and gravity

REVLib 2026

```
config.absoluteEncoder
    .apply(AbsoluteEncoderConfig.Presets
        .REV_ThroughBoreEncoderV2)
    .zeroOffset(ENCODER_OFFSET_ROT);

config.closedLoop
    .feedbackSensor(
        FeedbackSensor.kAbsoluteEncoder)
    .p(KP)
    .feedForward
    .kV(KV)
    .kCos(KG).kCosRatio(1.0);
```

Phoenix 6 2026

```
var feedback = config.Feedback;
feedback.FeedbackRemoteSensorID = ENCODER_ID;
feedback.FeedbackSensorSource =
    FeedbackSensorSourceValue.RemoteCANcoder;
feedback.SensorToMechanismRatio = 1.0;
feedback.RotorToSensorRatio = GEAR_RATIO;

config.Slot0.GravityType =
    GravityTypeValue.Arm_Cosine;
config.Slot0.kG = KG;
config.Slot0.kP = KP;
config.Slot0.kV = KV;
```

SAME JOB, BOTH VENDORS

Arm move: profile and command

REVLib 2026

```
config.closedLoop.maxMotion
    .cruiseVelocity(CRUISE_RPM)
    .maxAcceleration(ACCEL_RPM_PER_S)
    .allowedProfileError(PROFILE_ERROR_ROT);

public Command moveTo(double degrees) {
  double rotations = degrees / 360.0;
  return runOnce(() -> {
    target = rotations;
    controller.setSetpoint(rotations,
      ControlType.kMAXMotionPositionControl);
  }).andThen(waitUntil(this::atTarget));
}
```

Phoenix 6 2026

```
var magic = config.MotionMagic;
magic.MotionMagicCruiseVelocity = CRUISE_RPS;
magic.MotionMagicAcceleration = ACCEL_RPS2;

public Command moveTo(double degrees) {
  double rotations = degrees / 360.0;
  return runOnce(() -> {
    target = rotations;
    motor.setControl(
        request.withPosition(rotations));
  }).andThen(waitUntil(this::atTarget));
}
```

COMMON MISTAKE

Newest firmware is not always the right firmware

What you see	A SPARK shows “Outdated REV Device”, REVLib says the firmware is too old or too new, or Phoenix reports a firmware version error.
Why it happens	On both vendors, firmware and library are a matched pair for each season. The newest firmware listed can be for next season.
The fix	For 2026, pick the newest 26.x in Hardware Client 2, and the 2026 firmware year in Tuner X.

DECISION

How a young team can choose

Budget

Count whole systems. A Kraken includes its controller. A NEO needs a SPARK, and a Minion needs a Talon FXS.

What you own

Start from your shelf. KitBot SPARK MAX controllers, NEO V1.1 and Falcon 500 are all still legal.

Who can help

Choose what your team, or a team nearby, can debug at an event.

Mixing is allowed

R504 lists both vendors and one project can use both libraries. You pay with two apps and two update routines.

EXERCISE

Pick the parts for one mechanism

1. Choose one mechanism: an arm, a shooter or an intake.
2. Pick a motor and a controller. Check them on the pairing table.
3. Write down its current limit and where its sensor plugs in.
4. Open RevArm.java or CtreArm.java and find the gravity lines.

Done when: you have a motor, a controller, a current limit and a sensor, and have found the gravity lines.

NEXT SEASON

What changes for 2027

In 2026	Changing for 2027
SPARK firmware 26.x, conversion factors	Shipped already: firmware 27.0.0, conversion factors gone
Phoenix firmware for 2026	Alpha: Phoenix 26.70 needs 26.70.x firmware, no roboRIO
new TalonFX(id, bus) on a roboRIO	Alpha: int and String constructors go. CANBus takes a CANPort
new SparkMax(id, type)	Alpha: constructors take a WPILib CANPort
REV getters return plain values	Alpha: getters return a Signal, so call .get()
motor.set() on a Talon	Alpha: renamed setThrottle()

Firmware 27.0.0 is a normal REV release from 2026-10-29. Alpha items may change: REVLib 2027.0.0-alpha-7, Phoenix 6 26.70.0-alpha-2.

RECAP

What this deck covered

- Which motors work with which controllers.
- Why a 1 kHz loop on the controller beats a 50 Hz loop on the roboRIO.
- The same arm in REVLib and in Phoenix 6.
- What needs Phoenix Pro or a CANivore, and what does not.
- Choosing from budget, parts and available help.

Next: That ends the core course. Decks 8 to 10 go deeper.

REFERENCE

Official docs

These change every season, so check them before you buy.

SPARK closed loop

docs.revrobotics.com

REV Hardware Client 2

docs.revrobotics.com

Phoenix 6 free and Pro

v6.docs.ctr-electronics.com

Talon FXS motors

v6.docs.ctr-electronics.com